LANDFORMS OF GLACIATION / HIMALAYAN GLACIERS-GLOF

GLACIER BUDGET → RETREAT PARADOX → EROSION SUITE → FJORD-RIA FIREWALL → DEPOSITIONAL SORTING → ICE-MASS FAMILY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

★ PRIMARY CORE FLOW · ● SUBORDINATE EXTRA · ★ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

00 →

STAGE 00 GLACIER BUDGET

GLACIER BUDGET · SNOWFALL, AVALANCHE AND REFREEZING · ANNUAL GAIN EQUALS ANNUAL LOSS · MELT, SUBLIMATION AND CALVING WHERE RELEVANT · NET BALANCE · GEOMETRY RESPONDS WITH A LAG · MUST REMEMBER · GLACIATION PROCEEDS THROUGH ACCUMULATION-ABLATION BALANCE, ICE DEFORMATION/BASAL MOTION, EROSION

CORE CLAIM
- ACCUMULATION → snowfall, avalanche and refreezing
- ELA → annual gain equals annual loss

MECHANISM
- ABLATION → melt, sublimation and calving where relevant
- NET BALANCE → geometry responds with a lag

CONSEQUENCE / LIMIT
- MUST REMEMBER: Glaciation proceeds through accumulation-ablation balance, ice deformation/basal motion, erosion, transport and deposition
- cirque–arête–horn, U-shaped valley, moraine and outwash logic must connect to Himalayan mass balance, proglacial lakes and GLOF cascades.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Glaciation proceeds through accumulation-ablation balance, ice deformation/basal motion, erosion, transport and deposition; cirque–arête–horn, U-shaped valley, moraine and outwash logic must connect to Himalayan mass balance, proglacial lakes and GLOF cascades.

01 →

STAGE 01 RETREAT PARADOX

RETREAT PARADOX · ICE PARTICLES · CONTINUE MOVING DOWNSLOPE · RECEIVES FORWARD ICE FLUX · ABLATION DELIVERY · SNOUT SHIFTS UP-VALLEY · METRIC RULE · FLOW DIRECTION DIFFERS FROM MARGIN CHANGE

ICE PARTICLES → continue moving downslope → TERMINUS → receives forward ice flux → ABLATION DELIVERY → snout shifts up-valley → METRIC RULE → flow direction differs from margin change

CORE CLAIM
- ICE PARTICLES → continue moving downslope
- TERMINUS → receives forward ice flux

MECHANISM
- ABLATION DELIVERY → snout shifts up-valley
- METRIC RULE → flow direction differs from margin change

CONSEQUENCE / LIMIT
- ICE PARTICLES → continue moving downslope
- TERMINUS → receives forward ice flux

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: METRIC RULE → flow direction differs from margin change.

02 →

STAGE 02 EROSION SUITE

EROSION SUITE · JOINTED BLOCKS REMOVED · POLISH, STRIAE AND ROCK FLOUR · HEADWALL SYSTEM · HANGING VALLEY · VALLEY-SCALE OVERDEEPENING

PLUCKING → jointed blocks removed → ABRASION → polish, striae and rock flour → CIRQUE / ARETE / HORN → headwall system → U-TROUGH / HANGING VALLEY → valley-scale overdeepening

CORE CLAIM
- PLUCKING → jointed blocks removed
- ABRASION → polish, striae and rock flour

MECHANISM
- CIRQUE / ARETE / HORN → headwall system
- U-TROUGH / HANGING VALLEY → valley-scale overdeepening

CONSEQUENCE / LIMIT
- PLUCKING → jointed blocks removed
- ABRASION → polish, striae and rock flour

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CIRQUE / ARETE / HORN → headwall system.

03 →

STAGE 03 FJORD-RIA FIREWALL

FJORD-RIA FIREWALL · GLACIAL TROUGH LATER FLOODED BY SEA · RIVER VALLEY LATER DROWNED BY SEA · FJORD CUES · DEEP WATER, STEEP WALLS, HANGING VALLEYS · INUNDATION PRESERVES DIFFERENT PRE-EXISTING VALLEYS

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
FJORD → glacial trough later flooded by sea	FJORD CUES → deep water, steep walls, hanging valleys	FJORD → glacial trough later flooded by sea
RIA → river valley later drowned by sea	RULE → inundation preserves different pre-existing valleys	RIA → river valley later drowned by sea

CORE CLAIM
- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

MECHANISM
- FJORD CUES → deep water, steep walls, hanging valleys
- RULE → inundation preserves different pre-existing valleys

CONSEQUENCE / LIMIT
- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FJORD CUES → deep water, steep walls, hanging valleys.

04 →

STAGE 04 DEPOSITIONAL SORTING

DEPOSITIONAL SORTING · DIRECT ICE, GENERALLY UNSORTED · MELTWATER RIDGE IN OR UNDER ICE · BRAIDED MELTWATER, RELATIVELY SORTED · STREAMLINED DRIFT/TILL LANDFORM

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
TILL / MORAINE → direct ice, generally unsorted	OUTWASH → braided meltwater, relatively sorted	TILL / MORAINE → direct ice, generally unsorted
ESKER → meltwater ridge in or under ice	DRUMLIN → streamlined drift/till landform	ESKER → meltwater ridge in or under ice

CORE CLAIM
- TILL / MORAINE → direct ice, generally unsorted
- ESKER → meltwater ridge in or under ice

MECHANISM
- OUTWASH → braided meltwater, relatively sorted
- DRUMLIN → streamlined drift/till landform

CONSEQUENCE / LIMIT
- TILL / MORAINE → direct ice, generally unsorted
- ESKER → meltwater ridge in or under ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ESKER → meltwater ridge in or under ice.

05 →

STAGE 05 ICE-MASS FAMILY

ICE-MASS FAMILY · GLACIER, ICE CAP, ICE SHEET · SHELF, ICEBERG AND SEA ICE · SEA LEVEL · GROUNDED LOSS ADDS OCEAN MASS · SHELF LOSS CAN ACCELERATE INLAND DISCHARGE

GROUNDED → glacier, ice cap, ice sheet

FLOATING → shelf, iceberg and sea ice → SEA LEVEL → grounded loss adds ocean mass → BUTTRESSING → shelf loss can accelerate inland discharge

CORE CLAIM
- GROUNDED → glacier, ice cap, ice sheet
- FLOATING → shelf, iceberg and sea ice

MECHANISM
- SEA LEVEL → grounded loss adds ocean mass
- BUTTRESSING → shelf loss can accelerate inland discharge

CONSEQUENCE / LIMIT
- GROUNDED → glacier, ice cap, ice sheet
- FLOATING → shelf, iceberg and sea ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUTTRESSING → shelf loss can accelerate inland discharge.

06 →

STAGE 06 HIMALAYAN MAP HOOKS

HIMALAYAN MAP HOOKS · BARA SHIGRI · CHENAB SYSTEM · NUBRA-SHYOK-INDUS SYSTEM

GANGOTRI → Bhagirathi

ZEMU → Teesta → BARA SHIGRI → Chenab system → SIACHEN → Nubra-Shyok-Indus system

CORE CLAIM
- GANGOTRI → Bhagirathi
- ZEMU → Teesta

MECHANISM
- BARA SHIGRI → Chenab system
- SIACHEN → Nubra-Shyok-Indus system

CONSEQUENCE / LIMIT
- GANGOTRI → Bhagirathi
- ZEMU → Teesta

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BARA SHIGRI → Chenab system; SIACHEN → Nubra-Shyok-Indus system.

07 →

STAGE 07 WATER-TIMING SEQUENCE

WATER-TIMING SEQUENCE · EARLY LOSS · LOCAL MELT CONTRIBUTION MAY RISE · STORED ICE DECLINES · BUFFERING WEAKENS · LEAN SEASON · RELIABILITY RISK GROWS LOCALLY · MONSOON, SNOW AND GROUNDWATER VARY BY BASIN

EARLY LOSS → local melt contribution may rise → STORED ICE DECLINES → buffering weakens → LEAN SEASON → reliability risk grows locally → QUALIFIER → monsoon, snow and groundwater vary by basin

CORE CLAIM
- EARLY LOSS → local melt contribution may rise
- STORED ICE DECLINES → buffering weakens

MECHANISM
- LEAN SEASON → reliability risk grows locally
- QUALIFIER → monsoon, snow and groundwater vary by basin

CONSEQUENCE / LIMIT
- EARLY LOSS → local melt contribution may rise
- STORED ICE DECLINES → buffering weakens

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFIER → monsoon, snow and groundwater vary by basin.

08 →

STAGE 08 GLACIAL-LAKE TEST

GLACIAL-LAKE TEST · HOW WATER IS IMPOUNDED · SLOPE + ICE · POSSIBLE DISPLACEMENT TRIGGER · DAM CONDITION · OVERTOPPING, PIPING OR EROSION · EXPOSURE CONVERTS HAZARD INTO RISK

BASIN / DAM → how water is impounded

SLOPE + ICE → possible displacement trigger → DAM CONDITION → overtopping, piping or erosion → DOWNSTREAM → exposure converts hazard into risk

CORE CLAIM
- BASIN / DAM → how water is impounded
- SLOPE + ICE → possible displacement trigger

MECHANISM
- DAM CONDITION → overtopping, piping or erosion
- DOWNSTREAM → exposure converts hazard into risk

CONSEQUENCE / LIMIT
- BASIN / DAM → how water is impounded
- SLOPE + ICE → possible displacement trigger

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BASIN / DAM → how water is impounded.

09 →

STAGE 09 GLOF CASCADE

GLOF CASCADE · SLOPE FAILURE, INFLOW OR INSTABILITY · DAM EROSION BEGINS · SUDDEN RELEASE + SEDIMENT · CONSTRICTIONS AND ASSETS SHAPE CONSEQUENCES · CLOSE DISTINCTION · SNOWLINE, EQUILIBRIUM-LINE ALTITUDE, FIRN LINE AND GLACIER TERMINUS ARE

TRIGGER → slope failure, inflow or instability → WAVE / OVERTOPPING → dam erosion begins → BREACH → sudden release + sediment → VALLEY → constrictions and assets shape consequences → CLOSE DISTINCTION: Snowline, equilibrium-line altitude, firn line and glacier terminus are distinct

CORE CLAIM
- TRIGGER → slope failure, inflow or instability
- WAVE / OVERTOPPING → dam erosion begins

MECHANISM
- BREACH → sudden release + sediment
- VALLEY → constrictions and assets shape consequences

CONSEQUENCE / LIMIT
- CLOSE DISTINCTION: Snowline, equilibrium-line altitude, firn line and glacier terminus are distinct
- till is unsorted ice-laid sediment while outwash is meltwater-sorted, and a fjord is a drowned glacial trough rather than any steep inlet.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Snowline, equilibrium-line altitude, firn line and glacier terminus are distinct; till is unsorted ice-laid sediment while outwash is meltwater-sorted, and a fjord is a drowned glacial trough rather than any steep inlet.

10 →

STAGE 10 RISK-REDUCTION STACK

RISK-REDUCTION STACK · SATELLITE INVENTORY · CHANGING LAKES AND SLOPES · FIELD ASSESSMENT · DAM MATERIAL AND THRESHOLDS · WARNING + DRILLS · LAST-MILE ACTION · LAND USE + DESIGN

SATELLITE INVENTORY → changing lakes and slopes

FIELD ASSESSMENT → dam material and thresholds → WARNING + DRILLS → last-mile action → LAND USE + DESIGN → reduce exposure and fragility

CORE CLAIM
- SATELLITE INVENTORY → changing lakes and slopes
- FIELD ASSESSMENT → dam material and thresholds

MECHANISM
- WARNING + DRILLS → last-mile action
- LAND USE + DESIGN → reduce exposure and fragility

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Retreat varies by glacier and period
- lake growth alone does not prove imminent outburst, while dam material, freeboard, slope instability, ice/rock avalanches and downstream

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Retreat varies by glacier and period; lake growth alone does not prove imminent outburst, while dam material, freeboard, slope instability, ice/rock avalanches and downstream exposure condition risk.

11 →

STAGE 11 CRYOSPHERE ANSWER SPINE

CRYOSPHERE ANSWER SPINE · GLACIER, LAKE, GLOF AND RISK · GLACIER-BASIN RELATION · BUDGET, TRIGGER AND BREACH · HETEROGENEITY, ATTRIBUTION AND DATA DATE

DEFINE → glacier, lake, GLOF and risk → MAP → glacier-basin relation → EXPLAIN → budget, trigger and breach → QUALIFY → heterogeneity, attribution and data date

CORE CLAIM
- DEFINE → glacier, lake, GLOF and risk
- MAP → glacier-basin relation

MECHANISM
- EXPLAIN → budget, trigger and breach
- QUALIFY → heterogeneity, attribution and data date

CONSEQUENCE / LIMIT
- DEFINE → glacier, lake, GLOF and risk
- MAP → glacier-basin relation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPLAIN → budget, trigger and breach; QUALIFY → heterogeneity, attribution and data date.

E →

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · ABOUT 15,000 GLACIERS IN THE HIMALAYA, BETWEEN THE TWO · LOWEST LEVEL OF PERPETUAL SNOW; IT VARIES WITH LATITUDE · GREATER SNOWFALL/ACCUMULATION CAN LOWER THE CLIMATIC SNOWLINE, WHILE ARIDITY · MAIN GLACIERS LIE IN THE GREATER AND TRANS-HIMALAYA (KARAKORAM · THE LESSER HIMALAYA HAS ONLY SMALL GLACIERS (TRACES IN · SIACHEN LIES IN THE EASTERN KARAKORAM UNDER INDIAN CONTROL. · SOUTH LHONAK IS A MORAINE-DAMMED GLACIAL LAKE THAT EXPANDED

OPTIONAL SOURCE DEPTH
- Khullar: about 15,000 glaciers in the Himalaya, between the two syntaxial bends (east & west).
- Snowline = lowest level of perpetual snow; it varies with latitude, precipitation and local topography.
- Greater snowfall/accumulation can lower the climatic snowline, while aridity raises it; aspect,
- Main glaciers lie in the Greater and Trans-Himalaya (Karakoram, Ladakh, Zaskar)

ADVANCED DEBATE
- the Lesser Himalaya has only small glaciers (traces in Pir-Panjal, Dhauladhar).
- Siachen lies in the eastern Karakoram under Indian control. Biafo, Baltoro and Hispar are in the
- South Lhonak is a moraine-dammed glacial lake that expanded as its glacier retreated; use a named
- The October 2023 flood involved lake outburst/overtopping after intense precipitation and a

LEGACY / NUANCE
- Mitigation: glacial-lake monitoring, early-warning systems, and restricting development in the flood path (NDMA GLOF guidelines).
- Khullar's ~15,000 figure is a source-era Himalayan estimate; modern inventory counts are method-dependent.
- Snowline controls: latitude, precipitation, topography.
- More accumulation generally lowers snowline; exact north-south contrast varies by Himalayan sector.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels